



Do you have a passion for computing using new age technologies?

Do you want to work on leading-edge problems alongside some of the best & brightest in the world?

Do you like working in a dynamic working environment that involves creative problem solving and thinking on your feet?

We are in the lookout for the brightest young technologists of our generation to do their life's best work at NVIDIA.

We are hiring for our 6-month Jan-Jun Spring internship program across multiple positions in our Hardware group.

As an intern at NVIDIA, you will collaborate with a team of talented engineers tackling some of the most complex challenges in cutting-edge technology. Mentored by the best minds in the industry, you will have the opportunity to make meaningful contributions to exciting and impactful projects.

Here's a brief about the teams in our Hardware group -

GPU ARCHITECTURE TEAM

GPU architecture team is engaged in the development of industry leading high performance and power efficient GPUs.

Specific areas include architecture modelling, analysis, and performance verification. The team works on GPUs across all application domains such as gaming for PC and mobile devices, professional graphics and visualization and high-performance computation.

Skills you will use/develop:

- C++ modelling, test development
- TL design, debug
- ASIC design & verification tools, methodologies
- Computer architecture, Graphics, GPU micro-architecture, parallel computing
- Performance evaluation, analysis and debug
- Perl/Python scripting

Areas you will be working on:

- Computer Architecture,
- Memory Systems Architecture,
- Compiler Architecture / Performance Modelling

GPU ASIC DESIGN / VERIFICATION TEAM

Today NVIDIA's GPUs simulate human intelligence, running deep learning algorithms and acting as the brain of supercomputers, robots, and self-driving cars that can perceive and understand the world. We are seeking a passionate, innovative, and highly motivated senior verification engineer to join us in the development of the next generation of PCI Express controllers used in NVIDIA's GPUs and SOCs.

In this position, you will be responsible for verification of the ASIC design, architecture and microarchitecture using advanced verification methodologies. You are expected to understand the design and implementation, define the verification scope, develop the verification infrastructure and verify the correctness of the design. You will be working with architects, designers, pre and post silicon verification teams to accomplish your tasks.

What you'll be doing:

- Develop test plans, tests and verification infrastructure for PCIe at IP/sub system/SOC level
- Create verification environment using UVM methodology
- Create reusable bus functional models, monitors, checkers and scoreboards
- Drive functional coverage driven verification closure
- Work with architects, designers and post silicon teams

Ways to stand out from the crowd:

- Good knowledge of PCIe protocol Gen 3 and above
- Good debugging and problem-solving skills
- Good communication skills
- Ability/ desire to work as a team player

SoC TEGRA DESIGN & VERIFICATION

As a Hardware Engineer at NVIDIA, you will design and implement the industry's leading graphics, video and mobile communications processors. Specific areas include 2D and 3D graphics, mpeg, video, audio, network protocols, high-speed IO interfaces and bus protocols, and memory subsystem design. You will be responsible for Architecture and microarchitecture design of the ASICs, RTL design and synthesis, Logic and Timing verification using leading edge CAD tools and Semiconductor process technologies.

Areas you will be working on:

- ASIC, RTL, Design and Verification of Processors
- Low Power verification
- Power Estimation and Modelling
- Formal verification

CPU DESIGN & VERIFICATION TEAM

As a design and verification/validation engineer in the ARM CPU team, you will be working on the next generation of 64bit ARM CPUs and SOC. As part of this assignment, you will get a chance to learn about computer architecture at a very granular level, System Verilog, Design Verification, SOC Verification, Verification methodologies and C/C++ programming. You also will get an opportunity to get familiar with industry standard tools in verification and validation.

During the internship, you will contribute to building test benches, developing architectural simulators, modifying random instruction generators and creating stimulus for verification and validation of different units of the CPU and SOC.

Areas you will be working on:

- Computer Architecture
- Digital Design and Programming in C/C++/Perl
- ARM, CPU Design and Verification/ Validation

VLSI TEAM

VLSI team works in the areas of RTL Design, Verification, Design for Test, CAD tools, Synthesis, Timing, Place & Route, Circuit Design, Silicon Bring-up and Characterization. Responsible for state-of-the-art methodologies,



tools, flows and project execution on all Nvidia GPU, CPU, Auto, and Switch chips. As an intern you will be working on one or more such areas.

Skills you will use/develop:

- RTL Design, VCS, SV, UVM, Formal
- Verilog, C/C++, Python, TCL, Perl
- Logic Scan Test, Memory Test, High-speed IO Test, In-System Test
- Synthesis, Timing Closure (Primetime)
- Physical Design, Innovus, ICC2, Physical extraction, Place and Route, Floorplan
- SRAM, Analog, Digital circuit design, Hspice, EMIR, Silicon Correlation (ATPG, data visualization & analysis)
- Layout (Cadence Virtuoso)
- Silicon characterization for ageing, DPPM, Signal Integrity, Power Integrity, and familiarity with PC/SOC sub-system Architecture

INFRA TEAM

The Hardware Infrastructure Group supports all hardware and engineering development divisions for general cross-divisional development infrastructure. This group is responsible for a large breadth of existing and emerging tools supporting Engineering at NVidia.

What you'll be doing:

- Be a part of world class engineering teams to build reliable and maintainable software
- Work with your mentor to design well architected services that will run on top of sophisticated infrastructure
- Build solutions to continuously improve our High-Performance Computing (HPC) in cloud, on [1] prem and hybrid infrastructure
- Support Research & Development activities and engage in POCs/POVs for future improvements.
- Participate in tech communities, and interact with other members of the engineering community, both internal and external

What we need to see:

- Bachelor's or higher degree in Computer Science or Computer Engineering
- Proficiency with at least one of the following – Golang, Java, C++ and Python
- Excellent interpersonal skills with ability to simplify and communicate complex ideas to others.
- Excellent analytical and problem-solving skills with attention to detail
- Familiarity working in Linux environment.
- Proven strong software engineering skills through project contributions, internships or professional experience